Assignment 2 Report: Halfedge Mesh Traversal, Edge Flips, and Subdivision

Naufal Zafran Fadil
NPM 2406402542
CSCE604029 Computer Graphics, Universitas Indonesia

October 2026

Abstract

This report accompanies the halfedge mesh implementation (traversal, face geometry, mesh summary, edge flip, linear and Catmull–Clark subdivision). Section 1 describes the checks and gives mesh summaries before and after one subdivision step, Section 2 reports edge-flip acceptance and rejection, and Section 3 compares one to four steps of both subdivision schemes on the cube.

1 Checks and mesh summaries

All six unit-test groups pass. They check traversal on every provided mesh, boundary circulation, rotated face starts, normals and areas (including the L-shaped cap of area 8), rejected flips without mutation, double-flip reversal, and subdivision connectivity, vertex order, positions, and input immutability. The cube is also checked through four levels of each subdivision scheme.

Command: `python -m unittest -v test_student`

Each old edge adds one vertex and each old face adds one vertex, so $V' = V+E+F$. Each face corner becomes one quad, giving $F' = \sum_f \deg(f)$. Splitting each original edge contributes $2E$ edges; connecting each face point to its surrounding edge points contributes F', so $E' = 2E+F'$. Consequently $V' - E' + F' = V - E + F$, matching the table for both open and closed meshes.

2 Edge flip acceptance and rejection

Each edge index was tried independently on a freshly loaded mesh. Every accepted flip passed `mesh.validate()`, and a second flip restored each halfedge's original twin, next, target, edge, and face links. Every rejected flip left those links unchanged.

Commands: `python report_data.py`; `python -m unittest -v test_student`

Boundary edge. There is no second interior triangle. Applying the two-triangle rewiring would damage the boundary loop.

Nontriangle incident face. The six-halfedge local pattern no longer covers both face loops. Rewiring only those six halfedges would leave corners or edges disconnected from the intended faces.

Opposite vertices already joined. The new diagonal would duplicate an existing edge, so two halfedges would run between the same ordered vertices; `mesh.validate()` rejects that connectivity.

Table 1: Every provided mesh before and after one applicable subdivision step. Generated by `python report_data.py`. $\chi = V - E + F$ counts interior faces.

Mesh	Scheme	V	E	F	χ	Area
cube	input	8	12	6	2	24.000
cube	linear	26	48	24	2	24.000
cube	Catmull–Clark	26	48	24	2	11.528
fan	input	7	12	6	1	2.750
fan	linear	25	42	18	1	2.750
grid	input	16	24	9	1	9.125
grid	linear	49	84	36	1	9.112
lshape	input	12	18	8	2	23.200
lshape	linear	38	72	36	2	23.200
lshape	Catmull–Clark	38	72	36	2	14.106
octahedron	input	6	12	8	2	6.928
octahedron	linear	26	48	24	2	6.928
octahedron	Catmull–Clark	26	48	24	2	3.775
pyramid	input	5	8	5	2	11.211
pyramid	linear	18	32	16	2	11.211
pyramid	Catmull–Clark	18	32	16	2	3.903
tetrahedron	input	4	6	4	2	13.856
tetrahedron	linear	14	24	12	2	13.856
tetrahedron	Catmull–Clark	14	24	12	2	3.430
torus	input	72	144	72	0	12.823
torus	linear	288	576	288	0	12.823
torus	Catmull–Clark	288	576	288	0	11.219

Table 2: Independent attempts on every edge. Counts come from `python report_data.py`.

Mesh	Edges	Accepted	Boundary	Nontriangle	Already joined
octahedron	12	12	0	0	0
fan	12	6	6	0	0
pyramid	8	4	0	4	0
tetrahedron	6	0	0	0	6

3 Cube subdivision: one to four steps

Command: `python report_data.py` generates the counts, areas, valences, and sample renders below.

The schemes use identical quad connectivity, hence identical face and vertex counts: $F_k = 6 \cdot 4^k$ and $V_k = F_k + 2$. The eight original cube corners retain valence 3; newly inserted vertices have valence 4. Linear subdivision keeps old vertices fixed and places new points on the cube's planar faces, so the surface area remains 24. Catmull–Clark moves original and edge vertices inward, rounding the cube; its area falls to 9.229708 by step 4.

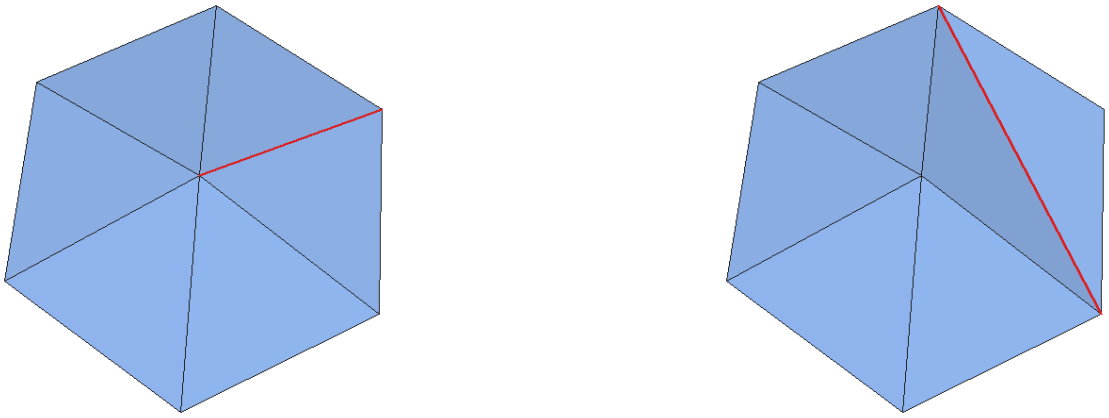


Figure 1: Fan edge 0 before (left) and after (right) its flip. The highlighted diagonal changes endpoints while the boundary stays intact. Both images are generated by `python report_data.py`.

Table 3: Cube face count, surface area, and number of vertices with valence different from four after each step. Vertex counts are included for reference.

Step	Faces	V	Linear area	Linear non-4	CC area	CC non-4
0	6	8	24.000000	8	24.000000	8
1	24	26	24.000000	8	11.527744	8
2	96	98	24.000000	8	9.727718	8
3	384	386	24.000000	8	9.326891	8
4	1536	1538	24.000000	8	9.229708	8

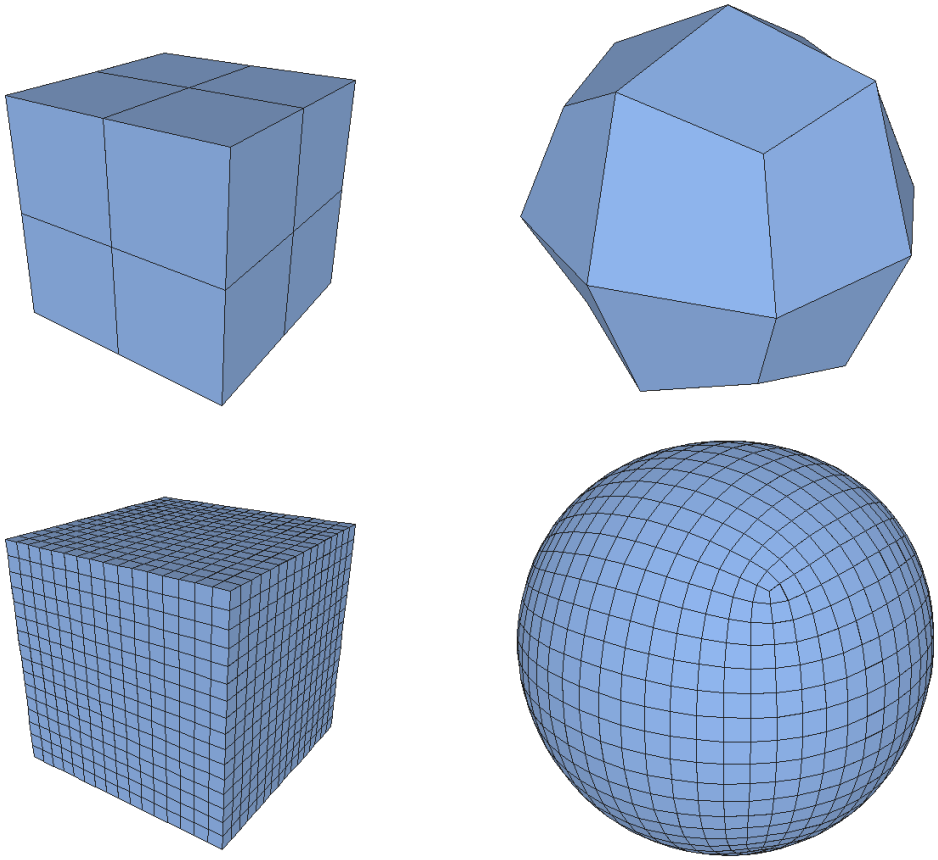


Figure 2: Linear (left) and Catmull–Clark (right) on the cube at steps 1 (top) and 4 (bottom). Generated by `python report_data.py`.